

9. Answer: C

When $a = 0.50 \text{ m s}^{-1}$,

$$F - f = (3.0 \times 10^5)(0.50)$$

$$F = 200 \text{ kN}$$

$$\therefore P = Fv = 2.0 \text{ MW}$$

Max speed occurs at $F_{\min}$, i.e. $F_{\min} = F$ at $a = 0$.

$$\therefore V_{\max} = \frac{P}{F_{\min}} = \frac{2.0 \times 10^6}{5.0 \times 10^4} = 40 \text{ m s}^{-1}$$

